

Virtual Private Cloud

Task-1. Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).

Steps

1. Log in to the **AWS Management Console**.
2. In the search bar at the top, type **VPC**.
3. Open **VPC**.
4. From the left menu, select **Your VPCs**.
5. Click **Create VPC**.
6. Under **Resources to create**, select **VPC only**.
7. Enter:
 - **Name tag:** InstaVibeVPC
 - **IPv4 CIDR:** 10.0.0.0/16
 - **IPv6 CIDR:** No IPv6 CIDR block
 - **Tenancy:** Default
8. Click **Create VPC**.

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

vpc-0dcbe34e1477fed50 / InstavibeVPC

Actions

Details

VPC ID vpc-0dcbe34e1477fed50	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0aa760d8315b09f13	Main route table rtb-01fb27c062ae8b773
Main network ACL acl-0293e91fd57943d6c	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 856727778179
Encryption control ID -	Encryption control mode -		

9.

Task-2. Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.0.0/24) for hosting a web server and one private subnet (e.g., 10.0.1.0/24) for backend services. Name your subnets as 'Public-Subnet' and 'Private-Subnet' for clarity.

Create Public Subnet

1. In the VPC Dashboard, select **Subnets**.
2. Click **Create subnet**.
3. Select your VPC: InstaVibeVPC.
4. Enter:
 - **Subnet name:** Public-Subnet
 - **Availability Zone:** Choose any available AZ, such as ap-south-1a.
 - **IPv4 subnet CIDR block:** 10.0.0.0/24
5. Click **Create subnet**.

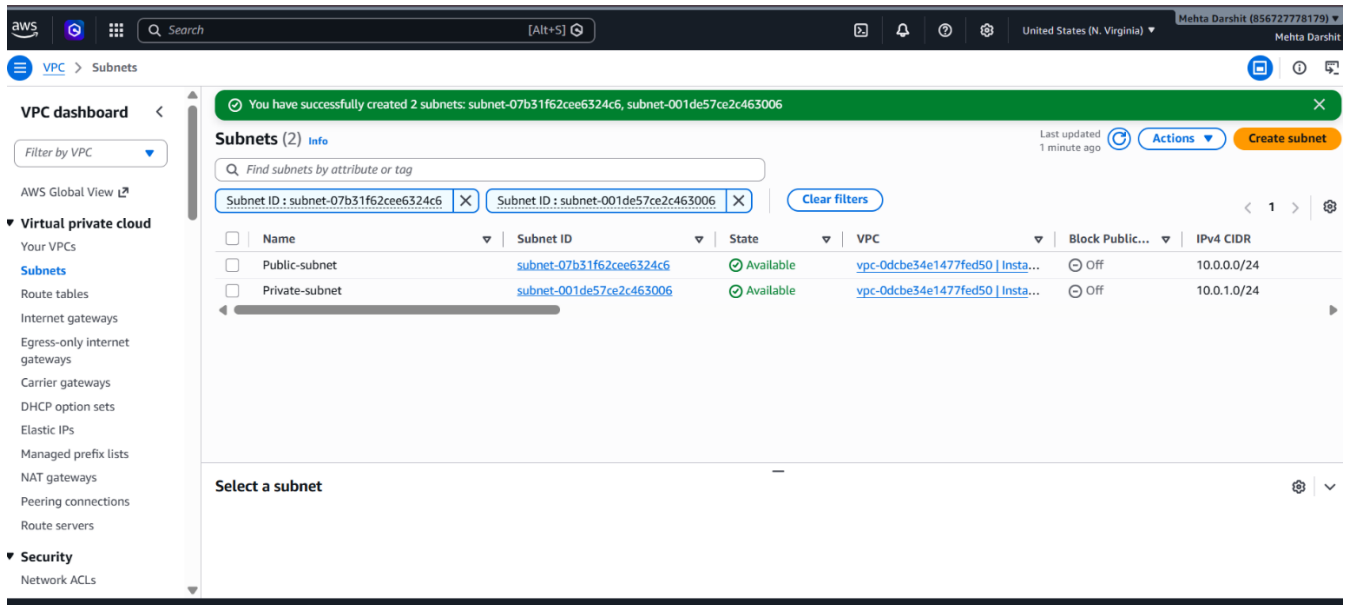
Create Private Subnet

Again select **Create subnet**.

Enter:

- **VPC:** InstaVibeVPC
- **Subnet name:** Private-Subnet
- **Availability Zone:** ap-south-1a
- **IPv4 subnet CIDR block:** 10.0.1.0/24

Click **Create subnet**.



Task-3. Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

Create Internet Gateway

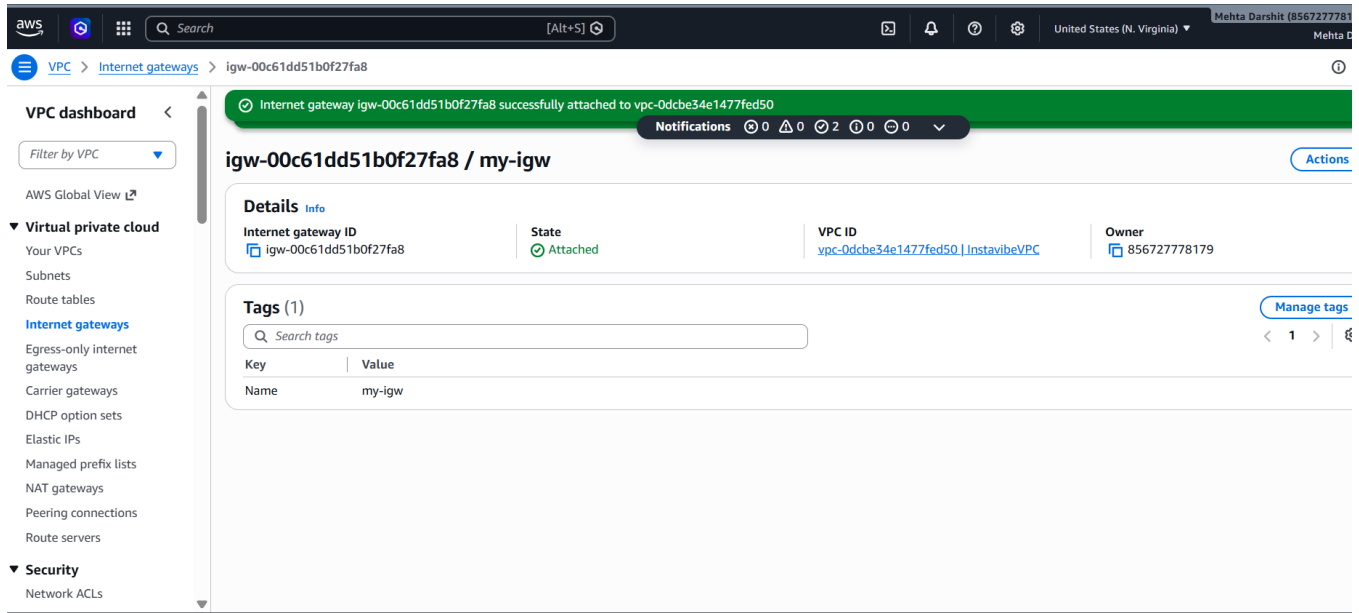
1. In the VPC Dashboard, select **Internet gateways**.
2. Click **Create internet gateway**.
3. Name it: my-igw
4. Click **Create internet gateway**.

Attach it to your VPC

1. Select my-igw
2. Click **Actions**.
3. Select **Attach to a VPC**.
4. Select:

InstaVibeVPC

5. Click **Attach internet gateway**.



Task-4 Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

A. Create Public Route Table

1. Go to **VPC → Route tables**.
2. Click **Create route table**.
3. Enter:

Name: Public-Route

VPC: InstaVibeVPC

4. Click **Create route table**.

Add Internet route

1. Select Public-Route
2. Go to **Routes**.
3. Click **Edit routes**.
4. Click **Add route**.
5. Enter:

Destination: 0.0.0.0/0

Target: Internet Gateway

6. Select my-igw
7. Save changes.

B. Associate Public Subnet

1. With Public-Route selected, go to **Subnet associations**.
2. Click **Edit subnet associations**.
3. Select:

Public-Subnet

4. Click **Save associations**.

Now the public subnet uses the Internet Gateway route.

C. Create Private Route Table

1. Go to **Route tables**.
2. Click **Create route table**.
3. Enter:

Name: Private-Route

VPC: InstaVibeVPC

4. Click **Create route table**.

The private route table should normally contain only the local VPC route:

Associate Private Subnet

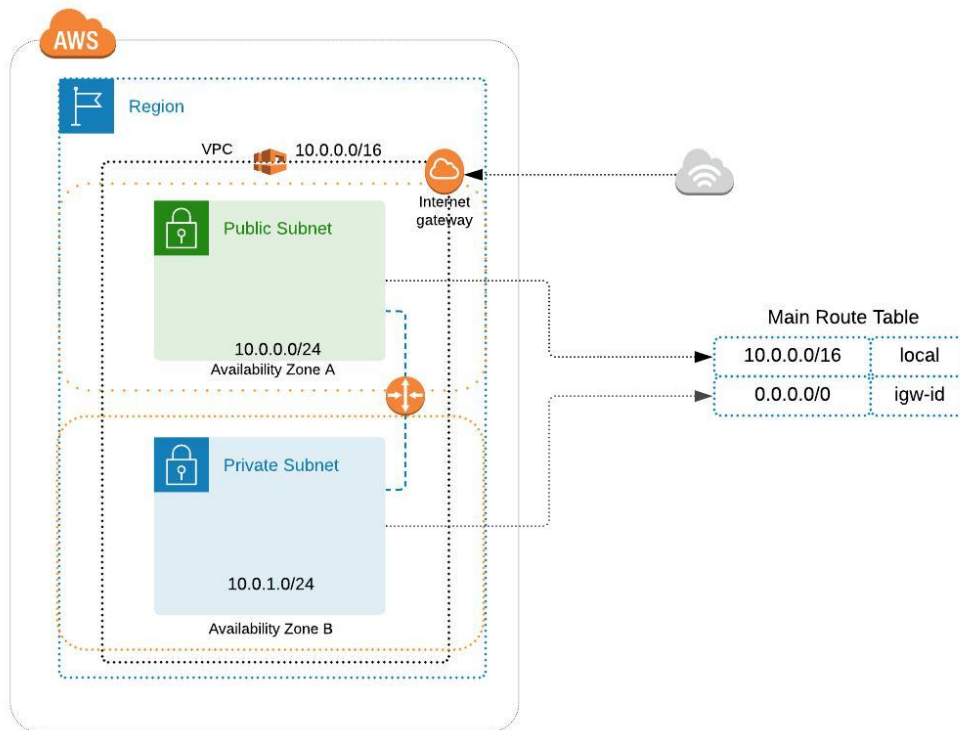
1. Select Private-Route
2. Go to **Subnet associations**.
3. Click **Edit subnet associations**.
4. Select:

Private-Subnet

5. Click **Save associations**.

The screenshot displays the AWS Management Console interface. On the left, the 'VPC dashboard' sidebar is visible, with 'Route tables' selected under the 'Virtual private cloud' section. The main content area shows the 'Route tables (1/4)' page. A table lists three route tables: a default route table, a 'public-route' (selected), and a 'private-route'. The 'public-route' is associated with the 'igw-00c61dd51b0f27fa8' Internet Gateway. Below this, the 'Routes (2)' section for the selected route table shows two routes: '0.0.0.0/0' pointing to the Internet Gateway and '10.0.0.0/16' pointing to 'local'. The bottom of the image shows a Windows taskbar with various application icons and system information.

5. Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.



Internet → Internet Gateway → InstaVibeVPC

Inside the VPC:

- Public-Subnet (10.0.0.0/24)
- Private-Subnet (10.0.1.0/24)

This VPC setup is similar to how an app like Zomato can separate its public-facing services from its private backend systems. The public subnet can contain web servers that receive requests from users through the Internet Gateway. The private subnet can contain databases and backend services that should not be directly accessible from the internet. This separation improves security by limiting direct internet access to sensitive backend resources.